

The Coordinates Are Not the Causal Object

Exact Future-Commitment Quotients in Deterministic Finite Agents

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Abstract

A recurring proposal in representation learning and mechanistic interpretability is that an internal vector, feature, neuron, or symbol is the scientifically primary object. A stronger alternative says that the invariant object is whatever internal structure preserves possible future commitments under intervention. We isolate the finite logical core of that proposal.

For deterministic finite Moore agents, we define two states as equivalent when every finite intervention word produces the same commitment. We prove that this relation is the greatest commitment bisimulation, is invariant under coordinate conjugacy, and induces the coarsest exact Markov abstraction for the registered intervention alphabet. A product-state search returns a finite distinguishing word whenever equivalence fails. These are specializations of classical automata minimization and bisimulation, not a novel quotient theorem.

We then execute a preregistered 2×2 construction over three delayed-commitment machine families, 64 fixed seeds, four coordinate/transition conditions, and 768 exact rows. Destroying coordinate identity and geometry while preserving transition and commitment conjugacy leaves future behavior unchanged. Preserving coordinates, current commitments, and every depth-one commitment while changing one delayed load-bearing transition changes the exact quotient. Leave-one-family-out balanced accuracy is 1.000 for quotient agreement and 0.500 for coordinate geometry, current output, and depth-one output.

The result is decisive but narrow. It proves that raw coordinates are neither necessary nor sufficient for future-commitment equivalence in the registered finite white-box class. It does not show that learned agents discover such quotients, that natural tasks admit the registered intervention family, or that constraint relaxation and repair can be automated. The formal experiment therefore sharpens the research target: the open problem is not defining the invariant after exhaustive access; it is learning the right intervention quotient and discovering which load-bearing constraint to revise.

1. The question that can actually be proved

The broad philosophical sentence

Intelligence preserves constraints across transformations.

is not a mathematical claim until constraint, transformation, and preserves are typed. The present paper deletes that ambiguity. We ask only:

When two finite internal states can be subjected to a fixed intervention alphabet, what state information is necessary and sufficient to predict every future commitment?

The answer is classical in mathematical form. It is the equivalence class induced by equality of future outputs, or equivalently the maximal output-preserving bisimulation. State aggregation and bisimulation have long used this principle to construct reduced decision processes [1--3]. Recent work gives a bounded-interaction Myhill--Nerode theorem for finite POMDP probe families, with a canonical, minimal, unique quotient [10]. Our deterministic finite result is a simpler specialization.

Why run an experiment if the theorem is known? Because the proposed representation ontology makes two empirical-sounding predictions that are often conflated:

1. radically different coordinates can realize the same operative state;
2. identical local coordinates and probes can conceal delayed consequences.

The registered factorial makes those predictions diverge exactly. It is a model-checking experiment on a constructed class, not a natural-data estimate.

2. Formal setup

2.1 Deterministic finite agents

An agent is a tuple:

$$A = (X, U, \text{delta}, \text{pi}, E)$$

where:

- X is a nonempty finite state set;
- U is a finite intervention alphabet;
- $\text{delta}: X \times U \rightarrow X$ is a total deterministic transition;
- $\text{pi}: X \rightarrow Y$ is a deterministic commitment label;
- $E: X \rightarrow R^d$ is an injective coordinate realization.

The coordinate map is intentionally separated from the transition/output object. It can be useful without being identifiable or invariant.

For w in U^* , let $\text{delta-star}(x, w)$ be the state after applying the finite intervention word w . The future-commitment signature is:

$$\text{Sigma}_A(x) = (\text{pi}(\text{delta}^*(x, w))) \text{ for every } w \text{ in } U^*$$

For states from possibly different agents with the same U and Y :

$$\begin{aligned} x \text{ ==_Sigma } y \\ \text{iff} \\ \text{pi}_A(\text{delta}_A^*(x, w)) = \text{pi}_B(\text{delta}_B^*(y, w)) \text{ for every } w \text{ in } U^* \end{aligned}$$

This is an interventional output equivalence. It is not an assertion that states have the same content, meaning, coordinate, history, or mechanism outside the registered interface.

2.2 Commitment bisimulation

A cross-agent relation R between X_A and X_B is a commitment bisimulation when

$$\begin{aligned} x R y \text{ implies } \text{pi}_A(x) = \text{pi}_B(y) \\ \text{and} \\ \text{delta}_A(x, u) R \text{delta}_B(y, u) \text{ for every } u \text{ in } U \end{aligned}$$

The quotient $X / \text{==_Sigma}$ is the paper's operative object. The term *constraint* refers only to the restrictions on future commitment traces captured by this quotient. No semantic ontology is inferred from a block.

3. Theorems

Theorem 1: future-commitment completeness

For finite deterministic agents with common typed alphabets, $x \text{ ==_Sigma } y$ if and only if x and y are related by the greatest commitment bisimulation.

Proof. Let R_{Sigma} relate states with identical signatures. Equality for the empty word gives equal current commitments. For any intervention u and continuation w , equality for the concatenated word uw

implies that the u -successors have equal signatures. Thus R_Sigma is a commitment bisimulation.

Conversely, if $x R y$ for a commitment bisimulation R , induction on word length gives related successor states after every common word. Related states have equal commitments, so their signatures agree. Therefore every commitment bisimulation is contained in R_Sigma , making R_Sigma the greatest one. QED.

Theorem 2: coordinate-gauge invariance

Let $g: X \rightarrow X'$ be a bijection satisfying:

$$\begin{aligned} g(\delta(x,u)) &= \delta'(g(x),u) \\ \text{and} \\ \pi(x) &= \pi'(g(x)) \end{aligned}$$

Then $x ==_Sigma g(x)$ for every x , and the two quotient machines are isomorphic.

Proof. Induction on word length gives $g(\delta\text{-star}(x,w)) = \delta\text{-prime-star}(g(x),w)$. Applying the output equality yields equal commitments for every word. The induced map on equivalence classes is therefore well-defined, bijective, output-preserving, and transition-preserving. QED.

Theorem 3: minimal exact Markov abstraction

Let $r: X \rightarrow Z$ admit a total abstract transition $\delta_bar: Z \times U \rightarrow Z$ and output $\pi_bar: Z \rightarrow Y$ such that:

$$\begin{aligned} r(\delta(x,u)) &= \delta_bar(r(x),u) \\ \text{and} \\ \pi(x) &= \pi_bar(r(x)) \end{aligned}$$

Then:

$$r(x) = r(y) \text{ implies } x ==_Sigma y$$

Hence any exact Markov abstraction refines the future-commitment quotient, and the quotient is the coarsest exact abstraction for the registered interface.

Proof. Equal abstract states remain equal under every abstract intervention word. Their abstract outputs, and therefore their ground outputs, agree after every word. The kernel of r is consequently contained in $==_Sigma$. The quotient map factors through r , proving coarseness. QED.

Theorem 4: finite distinguishing bound

If states in agents of sizes n_A and n_B are not equivalent, a distinguishing word exists with length strictly less than n_A times n_B .

Proof. Breadth-first search on the product state space starts from the state pair and explores one successor pair per intervention. If no pair with different outputs is reachable, the reachable-pair relation is a commitment bisimulation. If one is reachable, a shortest simple product path reaches it without repeating a pair. The product contains n_A times n_B pairs, so the path length is less than n_A times n_B . QED.

These theorems do not depend on the coordinate map E . They also do not identify which intervention alphabet should be used. That choice determines what the quotient preserves.

Theorems 1--4 are proved finite specializations or corollaries of classical automata-minimization, bisimulation, and distinguishing-sequence results; they are not presented as novel theorems.

4. Registered experiment

The preregistration was frozen at 15:14 EDT on 2026-07-27 before implementation or result inspection. Its SHA-256 digest is:

179afa61ba7052ab202533309bc9d4a74d3ea8d88a07ec2bc0008d02669e99c6.

4.1 Machine families

We construct three delayed-commitment Moore families:

- parity memory with two memory values;
- modulo-three memory with three values;
- order memory with none, last-zero, and last-one values.

Each family crosses memory with a three-phase clock. Outputs are defer until the terminal phase and then accept or reject. The common intervention alphabet is zero, one, advance, and reset.

The registered delayed mutant changes one phase-zero memory transition. It leaves the state set, coordinate rows, current commitments, clock transitions, and every depth-one commitment unchanged. The implementation rejects a mutant unless product search finds a later commitment witness.

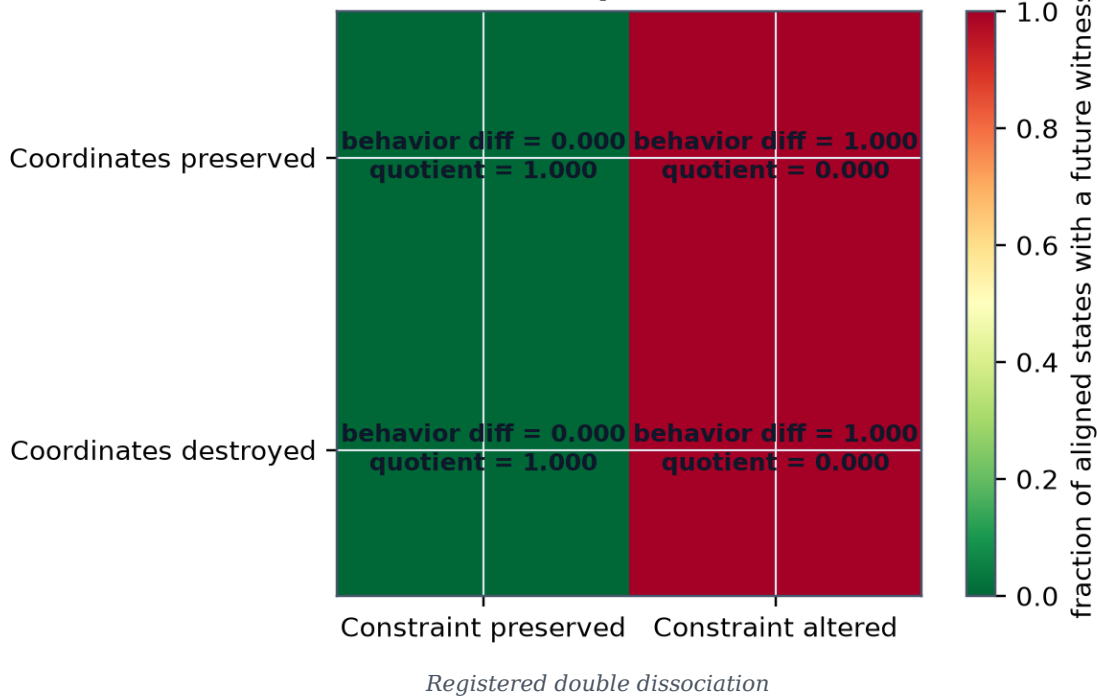
4.2 The 2×2 factorial

Cell	Coordinates	Transition constraint	Future behavior
RP-CP	preserved	preserved	predicted same
RD-CP	destroyed	conjugate	predicted same
RP-CA	preserved	one delayed edge altered	predicted different
RD-CA	destroyed	one delayed edge altered	predicted different

RP and RD mean representation preserved and destroyed. CP and CA mean future constraint preserved and altered.

For each seed, RD-CP and RD-CA share one nonidentity state permutation and one hash-seeded random injective coordinate assignment. This pairing makes coordinate statistics exactly independent of the constraint label. CP cells must satisfy output and transition conjugacy. CA cells must fail conjugacy and contain a delayed witness.

Behavior follows the future quotient, not coordinate identity



4.3 Baselines and exact target

We compare:

- aligned coordinate equality;
- correlation of Euclidean representational distance matrices;
- current-output agreement;
- depth-one successor-output agreement;
- exact quotient agreement.

The target is whether the full future constraint is preserved. Thresholds are fit on two families and tested on the third. Balanced accuracy is used because the current and depth-one baselines are constant.

This comparison is intentionally diagnostic. Quotient agreement is computed from the same exhaustive future behavior that defines the target. Its perfect score validates the formal bridge and implementation; it is not an empirical learning victory over tuned baselines.

5. Results

All 768 confirmatory rows were generated: 3 families $\times$ 64 seeds $\times$ 4 cells. All integrity and theorem-implementation gates passed.

5.1 Exact double dissociation

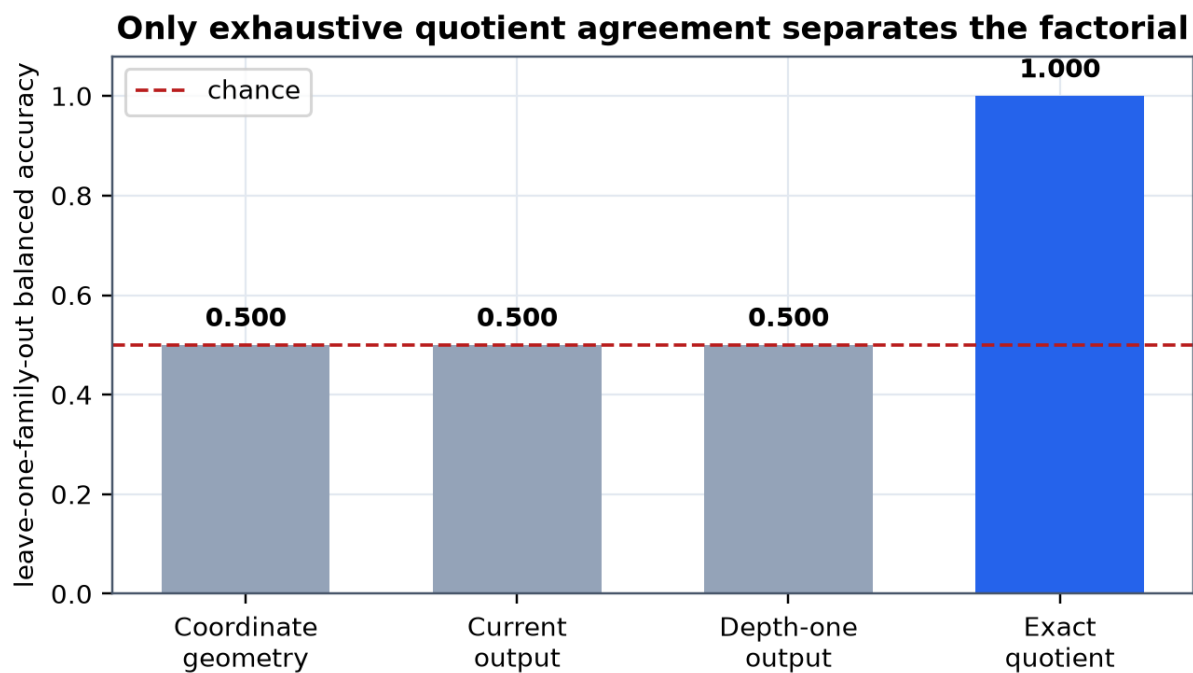
Cell	Coordinate equality	Geometry correlation	Depth-one agreement	Quotient agreement	Behavioral disagreement
RP-CP	1.000	1.000	1.000	1.000	0.000
RD-CP	0.000	-0.004	1.000	1.000	0.000
RP-CA	1.000	1.000	1.000	0.000	1.000
RD-CA	0.000	-0.004	1.000	0.000	1.000

The rows do not matter; the columns do. Destroying coordinates did not change behavior when the quotient was preserved. Preserving coordinates did not preserve behavior when the delayed transition constraint changed.

Every RD-CP state had no distinguishing word. Every RP-CA aligned state had a valid delayed witness of length at least two and below the product bound. The result replicated separately in all three families; no pooled score rescued a failed family.

5.2 Predictor separation

Predictor	Leave-one-family-out balanced accuracy
Coordinate geometry	0.500
Current output	0.500
Depth-one output	0.500
Exact quotient agreement	1.000



Predictor separation

The coordinate score is exactly nonpredictive because each coordinate realization appears once with the constraint preserved and once altered. Current and depth-one outputs are also matched by construction. The quotient separates the cells because it includes the delayed transition consequences that the local baselines omit.

5.3 Gate verdict

Gate	Result
F0 construction and provenance	PASS
F1 formal implementation	PASS
G1 representation non-necessity	PASS
G2 representation/local non-sufficiency	PASS
G3 factorial predictor separation	PASS
G4 family transfer	PASS
G5 claim calibration	PASS

Decision: ACCEPT_SCOPED_FINITE_QUOTIENT_CLAIM.

6. What has been proved

Within the registered deterministic finite class:

1. raw coordinates are not necessary for exact future behavior;
2. coordinate identity and depth-one outputs are not sufficient;
3. maximal commitment bisimulation is sufficient for the registered alphabet;
4. its quotient is the coarsest exact Markov abstraction for that interface;
5. inequivalence has a finite, machine-checkable intervention witness.

This gives precise content to one part of the phrase “representation as constraint preserved to commitment.” The operative invariant is not a particular vector. It is an equivalence class of states under all registered future commitment tests.

7. What has not been proved

The result does not establish the proposed major-ML capability. It withholds claims about learned constraint discovery, stochastic agents, real networks, natural tasks, and general intelligence.

7.1 The quotient was not learned

We had white-box transition tables and exhaustively refined the partition. Representation learning, causal abstraction learning, and interventional causal representation learning address the harder problem of identifying useful latent structure from finite observations or interventions [5, 7--9]. We solved none of those identification problems.

7.2 The intervention family was supplied

Equivalence is relative to U-star. A smaller probe family yields a coarser quotient; a richer family can split blocks. Nixon's bounded-interaction result makes this observer dependence explicit [10]. In a natural scientific or agentic setting, choosing the interventions is part of the discovery problem.

7.3 The target makes the quotient perfect by definition

The 1.000 predictor score is an exact consistency result, not an out-of-sample statistical surprise. It shows why local coordinates cannot substitute for the registered future object in this factorial. It does not show that an estimated quotient will beat CKA, probes, or causal abstractions on real networks. Representation similarity methods such as CKA are designed for different questions and invariance classes [4].

7.4 No constraint was discovered or repaired

The delayed mutant was authored by the experiment. The system did not infer a hidden assumption, rank candidate relaxations, or construct a minimal repair. Learned constraint discovery remains open.

7.5 Deterministic finite scope matters

Stochastic agents require distributional output equality and probabilistic bisimulation or a metric relaxation [1, 2]. Continuous state, partial observability, approximate equivalence, nonstationarity, and unbounded interaction introduce further identification and approximation choices. The present proofs cannot be exported unchanged.

8. Resolution of the earlier Constraint-Swap null

The earlier Constraint-Swap Causal Geometry experiment tested whether a hidden task constraint appeared as a reachability-aligned recurrent geometry and whether low-rank transports of that geometry selectively changed behavior. Competence and measurement-sensitivity controls passed, but

every registered geometry and causal gate failed.

This paper does not rescue that mechanism. It changes the object:

- Constraint-Swap tested learned hidden geometry in a recurrent agent.
- The present study computes an exact quotient from complete finite transition access.

These claims occupy different validity layers and evidence regimes. The null remains evidence against the geometry-mediated mechanism. The current positive is a formal construction showing what would be invariant if exhaustive future interventions were available.

9. Relation to prior work

Model minimization and bisimulation already establish coordinate-free state equivalence for planning and transition systems [1--3]. Bisimulation metrics relax exact equivalence to quantitative state similarity [2]. The Myhill--Nerode tradition characterizes minimal automata by future distinguishability, and the 2026 bounded-interaction theorem extends that logic to agent-limited POMDP probes [10].

Representation learning supplies complementary warnings. Unsupervised disentanglement is not identifiable without inductive biases [5], and underspecified pipelines can produce predictors with similar validation performance but different downstream behavior [6]. Causal abstraction and distributed alignment work use interventions to connect neural states to high-level causal models [7--9]. Our construction should be read as a minimal test fixture for those harder learning problems, not a replacement for them.

The contribution is therefore not a new equivalence relation. It is:

1. a preregistered 2×2 diagnostic separating coordinates from future constraints;
2. an exact bridge from representation language to classical quotient mathematics;
3. a calibrated conclusion that hidden-constraint discovery and repair remain open.

10. The next decisive experiment

The honest successor is not another coordinate scramble. It is a partial-access learning test:

1. hide the transition table;
2. expose a bounded training set of intervention traces;
3. require the system to choose informative interventions;
4. estimate an approximate quotient with calibrated uncertainty;
5. hold out one machine family and longer distinguishing words;
6. introduce several candidate delayed mutations, only one load-bearing;
7. require ranking the load-bearing mutation and proposing the smallest repair;
8. compare automata, bisimulation, causal, probe, geometry, and search baselines.

That would begin to test constraint discovery and representation repair. The current result supplies its oracle target and adversarial controls.

11. Conclusion

The finite formal question has a clean answer:

operative state = future-commitment equivalence class

for the stated deterministic finite agent, intervention alphabet, and exact commitment interface.

The experiment proves the intended double dissociation inside that scope. Coordinates can change while behavior does not; coordinates can stay fixed while delayed behavior changes. But the same result also blocks the grander claim from being presented as a new theory. The quotient is bisimulation in commitment language. The field-changing problem begins one level later: learning the quotient from partial access, discovering which constraint is load-bearing, and constructing a minimal repair that survives held-out domains.

References

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Reproduction

```
uv run --no-sync python -m \
  experiments.future_commitment_quotient.run_experiment
uv run --no-sync python scripts/make_future_commitment_quotient_figures.py
uv run --no-sync python scripts/build_future_commitment_quotient_pdf.py
uv run --no-sync python -m pytest -q \
  tests/test_future_commitment_quotient.py
```

Public exact rows: `experiments/future_commitment_quotient/results/registered_rows.jsonl`.

Public gate summary: `experiments/future_commitment_quotient/results/summary.json`.

The complete run payload remains under the gitignored `artifacts/future_commitment_quotient/` directory.